Hospital Billing and Patient Classification

1. Problem statement

Develop a Python program to calculate hospital bills using consultation, diagnostic, and treatment charges. Apply concessions based on patient category and classify the patient's payment status.

2. Objectives

- Calculate total hospital charges.
- Apply eligible concessions.
- Validate billing amounts.
- Classify patients based on the final bill.

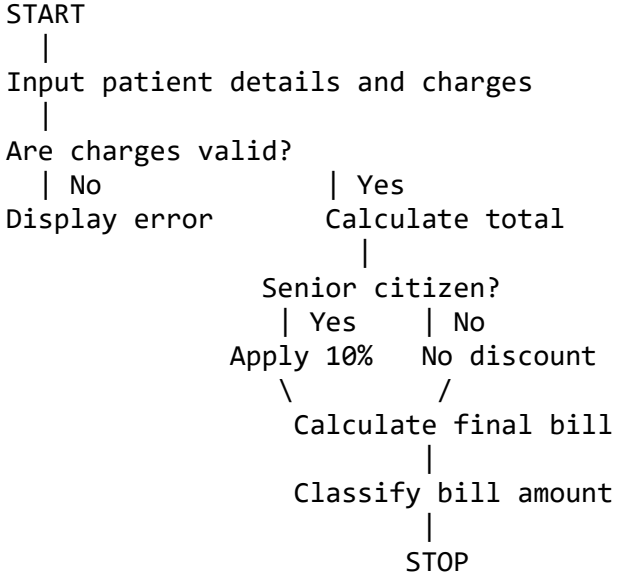
3. Assumptions

- Consultation charge: ₹500.
- Diagnostic charge and treatment charge are entered by the user.
- Senior citizens receive a 10% concession on the total bill.
- Final bill of ₹5,000 or more is classified as a high bill.
- Charges cannot be negative.

4. Algorithm

1. Input patient name, age, and charges.
2. Validate that charges are non-negative.
3. Calculate total charges.
4. Apply a 10% concession for senior citizens aged 60 or above.
5. Classify the bill.
6. Display the final bill and classification.

5. Flowchart



6. Pseudocode

```
START
INPUT name, age, consultation, diagnosis, treatment

IF any charge < 0
    PRINT "Invalid charges"
ELSE
```

```
    total = consultation + diagnosis + treatment
    IF age >= 60
        total = total * 0.90
    END IF

    IF total >= 5000
        classification = "High Bill"
    ELSE
        classification = "Regular Bill"
    END IF

    PRINT total, classification
END IF
STOP
```

7. Python source code

```
name = input("Enter patient name: ")
age = int(input("Enter patient age: "))
consultation = float(input("Consultation charge: "))
diagnostic = float(input("Diagnostic charge: "))
treatment = float(input("Treatment charge: "))

if min(consultation, diagnostic, treatment) < 0:
    print("Invalid charges")
else:
    total = consultation + diagnostic + treatment

    if age >= 60:
        total = total * 0.90

    print("Patient:", name)
    print("Final bill: Rs.", round(total, 2))

    if total >= 5000:
        print("Classification: High Bill")
    else:
        print("Classification: Regular Bill")
```

8. Test cases

Test	Age	Consultation	Diagnosis	Treatment	Expected result
1	25	500	1,000	2,000	₹3,500, Regular
2	65	500	2,000	4,000	₹5,850, High
3	40	500	2,000	4,000	₹6,500, High
4	30	-100	500	500	Invalid charges